VIT

Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

CHENNAI

Reg No

Final Assessment Test(FAT) - NOV/DEC 2025

Programme	**B.Tech.**	Semester	**Fall Semester 2025-26**
Course Code	**BCSE415L**	Faculty Name	**Prof. Sridhar R**
Course Title	**Human Computer Interaction**	Slot	**C1+TC1**
		Class Nbr	**CH2025260100551**
Time	**3 hours**	Max. Marks	**100**

Instructions To Candidates

- Write only your registration number in the designated box on the question paper. Writing anything elsewhere on the question paper will be considered a violation.

Course Outcomes

CO1: To design and develop processes and life cycle of human-computer interaction
CO2: To analyze product usability evaluations and testing methods
CO3: To apply the interface design standards/guidelines for cross-cultural and disabled users
CO4: To categorize, design and develop human-computer interaction in proper architectural structures

Section - I

Answer all Questions (4 × 15 Marks)

01. Meena, a postgraduate student, was attending her final online *viva* through the university portal. Due to heavy rain, her internet was unstable, and she was anxious when joining. On the screen, both the "Join Meeting" and "Submit Response" buttons appeared close together. Due to distraction, Meena inadvertently clicked "Submit Response" instead of "Join Meeting," which locked her out of the session. She missed her viva and was very upset, blaming both her stress and the confusing interface.
(a) Explain how Meena's emotional state influenced her perception, attention, and decision-making in this situation. (2 Marks)
(b) Identify the type of error committed by Meena. Justify your answer with reasons, and explain the possible causes of this error. (4 Marks)
(c) Suggest design improvements to the portal and justify them using Norman's seven principles of UX design. (9 Marks)

[15] (CO2/K4)

02. A user is booking a flight through an online system. To complete the task, the user must
- Read and compare several flight options.
- Remember preferred flight times, prices, and discount codes.
- Finally, select and confirm the booking.

During the booking process, the system briefly displays 8 random discount codes for 5 seconds. The user's working memory can hold only 5 items at a time. If the user does not rehearse, memory decays by 20% every 2 seconds. If the user rehearses, the decay reduces to 5% every 10 seconds. The recall test occurs after 10 seconds.
(a) Identify and explain the three main subsystems of the Model Human Processor (MHP) with a neat diagram. Describe their roles in the user's interaction with the flight booking interface (5 marks).
(b) Calculate how many digits (discount codes) the user can recall after 10 seconds without rehearsal and with rehearsal. Show your steps clearly. (5 marks)
(c) Analyze how working memory limits, decay time, and rehearsal influence recall in this task (2 marks)
(d) Suggest two design improvements to the flight booking interface that could help reduce cognitive load and improve user memory retention. Justify each improvement using MHP principles. (3 marks)

[15] (CO2/K4)

03. A university is developing an immersive Virtual Reality (VR) dashboard system for its public health and engineering departments. The system allows users to
- Interact with COVID-19 vaccination data in 3D, exploring trends across regions, age groups, and daily counts. Users can zoom, rotate, and manipulate the graphs in a virtual 3D space.
- Practice VR-based engineering tasks such as assembling complex machines in a virtual lab environment, using realistic motion and haptic feedback.

As the HCI designer, you are responsible for ensuring that VR statistical graphics are clear, informative, and easy to interpret.

You must also select VR input and output devices that allow users to interact naturally with 3D graphs and virtual objects while ensuring accuracy, comfort, and immersion.

(a). Explain the importance of well-designed statistical graphics in a VR environment for supporting data interpretation and decision-making. (4 marks)

(b). Recommend the most suitable types of statistical graphics for visualizing each of the following, and justify your choices using screen design and visual composition principles

i) Daily vaccination trends (3 marks)

ii) Comparison among regions (2 marks)

iii) Age-wise vaccination coverage (2 marks)

(c). Explain the role of devices in virtual reality and 3D interaction within HCI. Differentiate between input devices and output devices, giving suitable examples. (4 marks)

[15] (CO3/K5)

04. A user wants to open an existing document named report.docx, edit it, and save it as report_final.docx.
There are two methods available for completing the task:

Method 1:

- Move the mouse to the file icon on the desktop and double-click to open it.
- Once opened, use **File → Save As** from the menu.
- Type the new name report_final and press Enter.

Method 2:

- Open the application first by double-clicking its shortcut icon.
- From the **File → Open** menu, select the file report.docx.
- Then choose **File → Save As**, type the new name report_final, and press Enter.

Use the time estimates given in the table below (ignore system response time and thinking time):

Action	Explanation	Estimated Time
K	Keystroke (pressing a key)	0.28 s
T(N)	Type N characters	N × 0.28 s
P	Point with the mouse.	1.1 s
B	Press/release mouse button	0.1 s
BB	Double-click mouse button	0.2 s
H	Move hands to keyboard or mouse	0.4 s

(a) Identify and list the sequence of subtasks (using K, P, B, BB, H, and T(N)) for both Method 1 and Method 2. Represent each task as a clear action sequence. (3 marks)

(b) Using the given time estimates, calculate the total estimated execution time for each method. Show all your calculations clearly and present your results in a table. (6 marks)

(c) Based on your results, analyze and justify which method is more efficient, and explain how a change in interface design could reduce total task time. (2 marks)

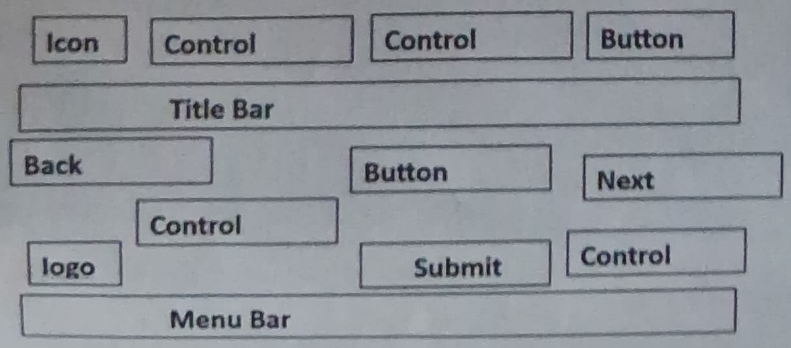
(d) Support your answer with Human Performance Modeling (KLM-GOMS) principles and explain how it can be applied to model user tasks in this scenario. (4 marks)

[15] (CO2/K3)

Section - II

Answer all Questions (4 × 10 Marks)

05. You are part of a design team developing the user interface for a desktop application that manages student records. During the prototype review, the screen layout was criticized for being cluttered and confusing, which is given below. Your supervisor asks you to redesign the interface so that users can easily find frequently used controls, identify navigation areas, and understand the logical flow of tasks.



(a) Identify and explain five major interface design problems present in the prototype screen elements. (5 marks)

(b) Recommend and justify five design improvements based on HCI screen design principles. Explain how each improvement will enhance usability and user efficiency in the redesigned student record system. (5 marks)

[10] (CO3/K4)

06. A UX team at HealthEase is developing a mobile health-tracking app that records daily steps, heart rate, and sleep data. Before final development, the team wants to test different interface designs and collect user feedback efficiently.

a) Apply your knowledge of prototyping to describe the steps involved in creating and testing a prototype for this health app. (6 Marks)

b) Evaluate whether a low-fidelity or high-fidelity prototype would be more appropriate at this stage. Justify your choice with reasons related to cost, user feedback, and design iteration. (4 Marks)

[10] (CO1/K5)

07. A university has launched a new Student Information Portal (SIP) that enables students to register for courses, check grades, pay fees online, and communicate with faculty. The development team seeks to ensure that the system is usable, efficient, and meets user expectations before deployment. As a Human–Computer Interaction (HCI) specialist, answer the following

a) Apply your understanding of usability and interface evaluation to identify and describe four suitable evaluation testing methods that can be used to validate the SIP's performance and user experience. *(8 marks)*

b) Evaluate and justify which one method or combination of validation methods would be most appropriate for this scenario, considering time, resources, and user involvement. *(2 marks)*

[10] (CO2/K3)

08. A mobile manufacturing company, TechNova, is planning to launch a new line of smartphones in emerging markets. The company must decide whether to use a licensed platform or an open-source platform. Additionally, TechNova aims to ensure that all layers of its mobile ecosystem, from network operators to application services, work efficiently to provide users with a seamless mobile experience. As a mobile technology consultant for TechNova, answer the following:

a) Analyze how each layer of the mobile ecosystem contributes to the performance, usability, and market success of TechNova's new smartphones. (7 Marks)

b) Evaluate and compare licensed vs. open-source platforms and recommend the best choice for TechNova with reasons. (3 Marks)

[10] (CO4/K4)

BL-Bloom's Taxonomy Levels - (K1-Remembering,K2-Understanding,K3-Applying,K4-Analysing,K5-Evaluating,K6-Creating)

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